

ModelSEED PubChem Cross-Validation

Production Run Results — May 2026

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Goal: Validate and correct 37,011 compound structures in ModelSEED against PubChem, and classify all discrepancies.

Pipeline (`Validate_PubChem_Structures.py`):

- Phase 0 Self-consistency — recompute formula/charge from InChI, flag SMILES $\leftrightarrow$ InChI mismatches
- Phase 1 External ID lookup — query PubChem via ChEBI/KEGG cross-references (highest confidence)
- Phase 2 Name lookup — batch compound name queries with multi-name disambiguation (for unresolved compounds)
- Phase 3 Recovery — direct InChIKey/InChI lookup in PubChem for compounds still unresolved after Phases 1–2
- Phase 4 Classification — InChIKey block comparison + cascading mismatch sub-classification
- Phase 5 Corrections — validate stereo (count + inversion check) and protonation (ChemAxon pK_a)
- Phase 6 Output — write corrected structure files, canonicalize SMILES, build Unique file using `List_ModelSEED_Structures.py` selection logic, record structural and formula

Metric	Count
Total compounds	37,011
Skipped (R-groups)	6,508
With InChIKey	30,431
With names	36,784
ChemAxon pKa loaded	45,708
Validated	30,333

Self-consistency fixes (Phase 0):

- Formula recomputed from InChI: 94
- Charge recomputed from InChI: 100
- SMILES parse failures: 12
- Total fixes persisted: 102

Validation scope: 30,333 compounds with both a searchable name and a stored InChIKey.

Validation Results (30,333 compounds)

Category	Count	%
MATCH	22,888	75.5
PROTON. DIFF	5,065	16.7
STEREO_DIFF	572	1.9
MISMATCH	759	2.5
NOT_FOUND	826	2.7
XREF_CONFLICT	155	0.5
AMBIGUOUS	68	0.2
Total	30,333	

- **94.0%** share the same connectivity with PubChem (MATCH + PROT + STEREO)
- **16.7%** PROTONATION_DIFF — same molecule, different charge state
- **1.9%** STEREO_DIFF — same connectivity, different stereochemistry
- **2.5%** MISMATCH — different molecular graph → sub-classified
- **2.7%** NOT_FOUND — no PubChem record

MISMATCH Sub-Classification (759 compounds)

Sub-category	Count
<i>Formula differs (non-H atoms):</i>	
wrong_mapping	353
salt_form_difference	3
<i>Formula differs (H only):</i>	
protonation_formula_diff	561
protonation_formula_diff_large	3
<i>Same formula — tautomer:</i>	
tautomer (RDKit canonical)	148
tautomer_connectivity (InChI)	178
tautomer_inchikey	3
ring_chain_tautomer	2
<i>Metal/coordination:</i>	
metal_representation_diff	66
<i>Needs manual review:</i>	
different_compound	2
likely_positional_isomer	3
cannot_parse_smiles	1

Cascading classification:

- 1 Formula comparison → wrong mapping or salt form
- 2 InChI layer analysis → protonation or stereo only
- 3 Ring/chain tautomer (O-ring count)
- 4 RDKit tautomer enumerator
- 5 Metal atom detection
- 6 InChI c-layer match → same heavy-atom scaffold
- 7 InChIKey stereo+protonation layers
- 8 Tanimoto (≥ 0.8) with positional isomer check

753 of 759 mismatches automatically classified as IGNORE.

Only 6 require manual review.

Corrections Overview (126 compounds, 651 field changes)

Correction	Count	What changes
Phase 0: Formula/charge consistency	102	Recompute formula and charge from InChI when stored values are inconsistent
Phase 5a: Stereochemistry	77	Replace SMILES, InChI, InChIKey with PubChem's (more stereocenters defined, no inversions)
Phase 5a: Stereo pH 7 normalization	20	Adjust accepted stereo corrections to ChemAxon target charge (prevents neutral PubChem forms from overwriting curated ionization)
Phase 5b: Protonation (PubChem + pKa)	23	For PROTONATION_DIFF compounds: adjust stored SMILES charge when ChemAxon pKa confirms PubChem
Phase 5c: Protonation (pKa only)	26	Independent of PubChem: adjust any compound whose stored charge disagrees with ChemAxon pKa at pH 7
Phase 6: SMILES canonicalization	86	Rewrite non-canonical SMILES in RDKit canonical form

Fields changed per compound: SMILES, InChI, InChIKey, and (where applicable) formula and charge.

Acceptance criteria (all must hold):

- 1 PubChem defines $\geq$ stereocenters than stored
- 2 Shared stereocenters verified via InChI /t (tetrahedral) and /b (double-bond) layers — no inversions allowed
- 3 PubChem SMILES/InChIKey internally consistent

pH 7 normalization (20 of 77):

- PubChem stores neutral forms
- Before applying, adjust PubChem SMILES to ChemAxon target charge at pH 7
- Prevents overwriting curated ionization states

Rejection reasons (1,254):

- PubChem has **fewer** defined stereocenters than stored (360)
- Stereo **inversions** at shared centers (135)
- MISMATCH sub-type: wrong mapping, tautomer, or different compound (759)

Key design choice: only *add* missing stereochemistry, never change existing assignments. Conservative approach avoids overwriting curated data from literature sources.

5b. PubChem + pKa (23 corrections):

- Compound classified as PROTONATION_DIFF
- ChemAxon pK_a confirms that the stored charge is wrong at pH 7
- Adjusts *stored* SMILES to target charge (does **not** replace with PubChem's neutral form)

5c. pKa only (26 corrections):

- Runs on **all** 30,333 compounds
- Independent of PubChem comparison
- If stored charge $\neq$ ChemAxon target at pH 7 and no borderline pKa $\rightarrow$ correct

Safety checks (both paths):

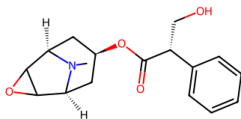
- Borderline pK_a (6.0–8.0) $\rightarrow$ flagged, not corrected (5 compounds)
- Cross-validation: if ChemAxon ionic charge is closer to stored than to target $\rightarrow$ skip (90)
- After adjustment: verify InChIKey connectivity unchanged (protonation-only change)

Design choice: use ChemAxon pK_a as the authority for pH 7 ionization, not PubChem (which stores neutral forms inappropriate for metabolic modeling).

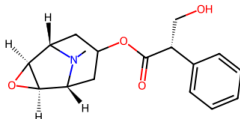
Example: Stereochemistry Corrections Applied

cpd01274 — Scopolamine

cpd01274 [STEREO_DIFF]
ModelSEED



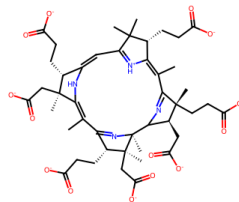
PubChem CORRECTED



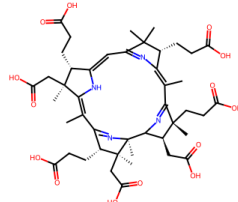
Tropane alkaloid: PubChem adds missing stereocenters on the epoxide and ester moieties.

cpd03832 — Hydrogenobyrrinate

cpd03832 [STEREO_DIFF]
ModelSEED



PubChem CORRECTED



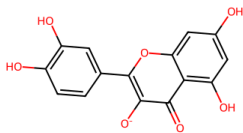
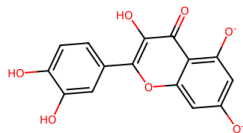
Corrinoid: PubChem defines additional stereocenters on the macrocyclic ring system.

Example: Protonation (pKa) Corrections Applied

cpd00313 — Quercetin

cpd00313 [PROTONATION_DIFF]
ModelSEED

PubChem CORRECTED

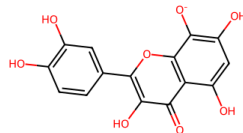
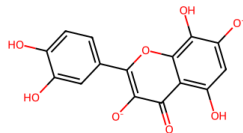


Flavonoid: two phenolate [O⁻] groups protonated to OH based on ChemAxon pK_a. Charge: -2 → 0. Formula: C₁₅H₈O₇ → C₁₅H₁₀O₇.

cpd02539 — Gossypetin

cpd02539 [PROTONATION_DIFF]
ModelSEED

PubChem CORRECTED



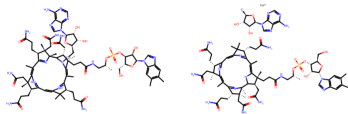
Hexahydroxyflavone: two phenolate [O⁻] groups protonated to OH. Charge: -2 → 0. Formula: C₁₅H₈O₈ → C₁₅H₁₀O₈.

Example: Discrepancies Not Corrected

Protonation formula diff

cpd00166 [MISMATCH]
ModelSEED

PubChem
REJECTED

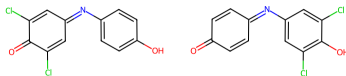


cpd00166 (Cobalamin): ModelSEED and PubChem differ by H count only ($H_{diff} = -2$). Stored protonation state preserved for metabolic modeling.

Tautomer ignored

cpd00088 [MISMATCH]
ModelSEED

PubChem
REJECTED

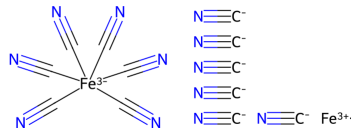


cpd00088: classified `tautomer_connectivity` — same InChI c-layer, different H positions. Stored form curated for biological relevance.

Metal repr. ignored

cpd00271 [MISMATCH]
ModelSEED

PubChem
REJECTED



cpd00271 (Ferricyanide): connected vs. disconnected Fe-CN SMILES. Same compound, different convention.

File	Rows	Description
<i>Corrected structures (Biochemistry/Structures/):</i>		
All_ModelSEED_Structures_updated.txt	281,436	All structure records (multi-row per cpd)
Unique_ModelSEED_Structures_updated.txt	97,607	Unique charged structures (consistent with List_ModelSEED_Structures.py; excludes formula conflicts)
<i>Reports (Scripts/Structures/structure_review_output/):</i>		
pubchem_validation_report.tsv	30,333	One row per compound with result type
pubchem_mismatches.tsv	1,323	MISMATCH entries with sub-classification
pubchem_protonation_diffs.tsv	5,065	Protonation-only differences
pubchem_stereo_diffs.tsv	572	Stereochemistry-only differences
pubchem_tautomer_diffs.tsv	331	All tautomer sub-types combined
pubchem_rejected_corrections.tsv	1,254	Corrections that failed validation
pubchem_corrections_log.tsv	—	Per-field correction log
pubchem_protonation_corrections.tsv	—	Protonation correction details (pKa)
pubchem_xref_conflicts.tsv	—	ChEBI/KEGG cross-ref CID conflicts
pubchem_smiles_failures.tsv	—	Compounds with unparseable SMILES
pubchem_stereo_review.tsv	—	Single-inversion stereo cases for review
consistency_report.tsv	242	Phase 0 self-consistency issues

Unique Structure Selection (`--rebuild-unique`)

Algorithm (consistent with `List_ModelSEED_Structures.py`):

- 1 Filter curated/ignored aliases (Curation/*.txt, 135 entries)
- 2 Priority: Charged InChI > Original InChI > Charged SMILE > Original SMILE
- 3 Formula conflict → **exclude** compound
- 4 Multi-database consensus: prefer structures appearing in ≥ 2 databases
- 5 Source priority fallback: MetaCyc > KEGG > ChEBI > Rhea
- 6 InChIKey connectivity analysis with stereochemistry preference

Updated Unique file statistics:

- 36,883 unique compounds (97,607 rows)
- 42 compounds newly included: corrections resolved formula conflicts
- 3 compounds excluded: unresolved formula conflicts (cpd02744, cpd03148, cpd19421)
- 2,449 structural conflict entries recorded
- 418 formula conflict entries recorded

Standalone usage:

```
python Validate_PubChem_Structures.py  
--rebuild-unique
```

Rebuilds Unique file from All file without rerunning the validation pipeline.

- ① **Manual review** (6 compounds):
 - 2 genuinely different compounds (cpd46969, cpd32805)
 - 3 likely positional isomers (cpd22821, cpd23345, cpd26545)
 - 1 invalid stored SMILES (cpd46870)
- ② **NOT_FOUND** (826) — no PubChem record; mostly rare metabolites or generic class names
- ③ **XREF_CONFLICT** (155) — ChEBI/KEGG map to different PubChem CIDs
- ④ **AMBIGUOUS** (68) — compound names map to multiple PubChem CIDs
- ⑤ **Edge cases**: 2 antimycin positional isomers (cpd47754/cpd47782) labeled `tautomer_inchikey` — InChI c-layer cannot distinguish bond-order-only differences in acyl chains